Generic Accessibility Completion

Incidence Varieties, Rank-Protected Bridges, and the Generic Completion Principle

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This paper is Part VII of the RIME program. Paper IV gives the collision geometry, Paper V gives the local repair calculus, and Paper VI gives the deformation geometry of accessibility walls. Paper VII isolates the generic completion boundary: outside high-codimension incidence, rank-protected bridge products are expected to survive, giving the conjectural route from (R_1, R_2) to the first-depth invariant D.

Abstract

Problem. Why is accessibility generically stable? Paper V identifies local length-two obstruction mechanisms, and Paper VI places the corresponding accessibility data over a deformation base. The remaining static question is the completion problem: when does the accessibility jet

$$\mathcal{J}_{\text{acc}} = (J_{\text{block}}, J_{\text{comm}}, J_{\text{depth}})$$

determine the accessibility depth matrix D?

Approach. The paper combines three blocks. Paper V supplies the local mechanism taxonomy. Paper VI supplies the wall hierarchy

$$\widehat{\Sigma}_{R_1} \subseteq \widehat{\Sigma}_{R_2} \subseteq \widehat{\Sigma}_D = \Sigma_{\text{access}} \subseteq \Sigma_{\text{spec}} \subseteq \Sigma_{\text{comm}}$$

and interprets R_1, R_2, D as discrete shadows of $\mathcal{J}_{\text{acc}}$. The new input developed here identifies Type IV incidence as a high-codimension algebraic degeneration:

$$I = \{(A, B) : AB = 0,\ A \neq 0,\ B \neq 0\}.$$

For $A \in \mathbb{C}^{m\times n}$, $B \in \mathbb{C}^{n\times p}$, and $\operatorname{rank} A = r$, the fixed-rank incidence stratum has

$$\operatorname{codim} I_r = (m - r)(n - r) + pr.$$

For square $d \times d$ blocks, the dominant incidence component has $r \approx d/2$ and codimension asymptotic to $3d^2/4$, or about 37.5% of the ambient dimension $2d^2$.

Results. The theorem-level result is the codimension calculation for the Type IV incidence variety. The completion statement is formulated as a conjectural theorem program:

> **Conjecture 1 (Completion Away from Incidence).** Under the stated richness and nondegeneracy assumptions, on the generic non-incidence locus where all bridge products avoid $AB = 0$, the accessibility jet $\mathcal{J}_{\text{acc}}$ has a locally constant discrete shadow, and that shadow determines the accessibility depth matrix D.

Computational support is consistent with this program. Synthetic Type III systems lie inside the candidate completeness class; synthetic Type IV systems lie outside it; perturbation at $\varepsilon = 10^{-6}$ destroys Type IV in 100/100 trials for the tested incidence models; 0/400000 random matrix pairs satisfy $AB = 0$ with $A, B \neq 0$; and exact-hash audits on 80 diverse random systems found zero disagreements among systems with identical (R_1, R_2). A claim-status-gated rank-protected bridge audit further separates the generic and structured regimes: five fixed-seed

random systems give 1080/1080 safe bridge products and zero incidence candidates, while the Rubik system has 528 bridge-level incidence candidates concentrated in a small family of sector triples. An exploratory ablation over dimension, rank-deficient fraction, and generator count found zero incidence candidates across 75 random runs and 23400 audited bridge products.

Implications. Paper V identifies local exceptional mechanisms. Paper VI places those mechanisms on the generator-set moduli space. Paper VII explains why the hardest mechanism, Type IV incidence, is nongeneric: it lies on a high-codimension algebraic variety. The resulting principle is conditional but sharp: away from incidence and under richness/nondegeneracy hypotheses, the accessibility jet is expected to complete the first-depth calculation. Rubik is therefore not a random-generic system; it is a stable carrier of high-codimension algebraic structure.

Notation Table

Symbol	Meaning
V	finite-dimensional complex Hilbert space
$\mathcal{S} = (V, \{Q_i\}, \mathcal{X})$	sectorized observable framework
Q_i	orthogonal sector projector, $\sum_i Q_i = I$
X_g	skew-Hermitian generator
$B_{ij}^g = Q_i X_g Q_j$	projected generator block
$R_1(i, j; g)$	generator support: 1 iff $B_{ij}^g \neq 0$
$R_2(i, j; g, h)$	projected commutator survival: 1 iff $Q_i[X_g, X_h]Q_j \neq 0$
$D(i, j)$	minimal Lie depth connecting sector j to sector i
$\mathcal{J}_{\mathrm{acc}}$	accessibility jet $(J_{\mathrm{block}}, J_{\mathrm{comm}}, J_{\mathrm{depth}})$
Σ_{comm}	outer commutativity locus $\{w : [Q_T(w), H_T(w)] = 0\}$
Σ_{spec}	normal spectral chart inside Σ_{comm} where joint projectors and accessibility fields are used
Σ_{IV}	Type IV incidence locus
C	candidate generic completeness class, excluding Type IV incidence

Depth convention: depth 0 means a direct generator block, depth 1 means a projected commutator block, depth 2 means a nested commutator, and $D(i, j) = \infty$ means frozen in the tested Lie filtration.

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1 Completion Problem

Why is accessibility generically stable? The answer proposed here is that the hard local obstruction, Type IV incidence, is a high-codimension algebraic condition. Away from this incidence locus, rank-protected bridge products survive in the generic matrix model, so the accessibility jet is expected to complete the first-depth calculation under the stated richness hypotheses.

Paper V established the local accessibility calculus for a fixed sectorized observable framework

$$(V, \{Q_i\}, \{X_g\}).$$

The first layer R_1 records which generator blocks are nonzero. The second layer R_2 records which projected commutators survive. Paper V showed that R_1 is not enough: binary support sees possible paths, but it does not see whether projected matrix products cancel.

Paper VI then moved the same observables over normal spectral charts $\Sigma_{\mathrm{spec}} \subseteq \Sigma_{\mathrm{comm}}$. There, $R_1(w)$, $R_2(w)$, and $D(w)$ are discrete rank/support shadows of smooth matrix fields. The accessibility wall hierarchy records where these shadows fail to continue locally.

The remaining question is whether the accessibility jet determines D.

Equivalently, when is the data visible through R_1, R_2, and the Hall-propagation component of $\mathcal{J}_{\mathrm{acc}}$ complete for the first-depth matrix?

This paper does not claim an unconditional theorem that (R_1, R_2) determines D. That statement is false without hypotheses. Instead, it isolates the generic obstruction boundary and formulates the completion theorem on the non-incidence locus.

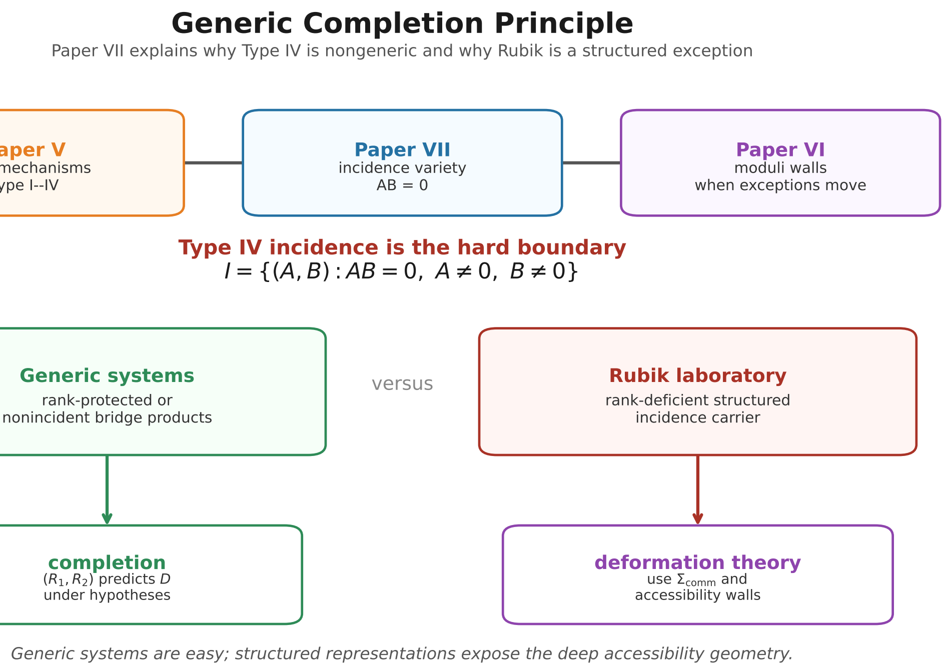

Figure 1. Paper VII overview. The Generic Completion Principle predicts completion away from high-codimension incidence, supported by rank-protected bridge survival. Rubik sits in a structured exceptional region that is described by incidence geometry and connected back to the deformation theory of Paper VI.

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2 Local Mechanisms

Paper V gives a length-two mechanism taxonomy. For sector pair (i, j) and generators g, h, expand

$$Q_i[X_g, X_h]Q_j = \sum_k \left(Q_i X_g Q_k X_h Q_j - Q_i X_h Q_k X_g Q_j \right).$$

The local mechanisms are:

Type	Mechanism	Status
Type I	singleton-color degeneracy	no depth-one repair
Type II	projected commutator survives	repaired at R_2
Type III	signed cancellation among nonzero products	exceptional but soft
Type IV	termwise vanishing by image-kernel incidence	exceptional and hard

Type III is a relation among products that exist. Type IV is different: the candidate products vanish before signed cancellation can occur. For a single bridge product this means

$$A = Q_i X_g Q_k, \qquad B = Q_k X_h Q_j, \qquad AB = 0$$

with $A \neq 0$ and $B \neq 0$, equivalently

$$\operatorname{im} B \subseteq \ker A.$$

Thus Type IV is an incidence condition, not merely a cancellation condition.

Definition 2.1 (Candidate Completeness Class)**.**
Let C denote the class of sectorized systems whose length-two $R_2 = 0$ obstructions are Type I or Type III, with no Type IV incidence obstruction.

This is a candidate generic completeness class, not a universal theorem class. The generic completion conjecture below states that, after excluding incidence and assuming the usual nondegeneracy/richness hypotheses, the accessibility jet should determine D.

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3 Type IV Incidence Variety

The main algebraic input of Paper VII is the codimension of the Type IV incidence condition.

Theorem 3.1 (Incidence Codimension)**.**
Let

$$A \in \mathbb{C}^{m\times n}, \qquad B \in \mathbb{C}^{n\times p},$$

and let

$$I_r = \{(A,B) : AB = 0,\ \operatorname{rank} A = r\}.$$

Then

$$\dim I_r = r(m+n-r) + np - pr$$

and the codimension of I_r in the ambient space $\mathbb{C}^{m\times n} \times \mathbb{C}^{n\times p}$ is

$$\operatorname{codim} I_r = (m-r)(n-r) + pr.$$

Proof. *The variety of $m \times n$ matrices of rank r has dimension $r(m+n-r)$. For fixed rank-r matrix A, the equation $AB = 0$ forces the columns of B to lie in $\ker A$, whose dimension is $n-r$. Hence the fiber over A has dimension $p(n-r) = np - pr$. Adding base and fiber dimensions gives the stated dimension. Subtracting from the ambient dimension $mn + np$ gives*

$$mn + np - (r(m+n-r) + np - pr) = (m-r)(n-r) + pr.$$

This proves the formula.

Corollary 3.2 (Square Blocks)**.**
For square blocks $m = n = p = d$,

$$\operatorname{codim} I_r = (d-r)^2 + dr.$$

For the Type IV locus, the nonzero/non-rank-protected square ranks are $1 \le r \le d-1$. The dominant component among these strata minimizes this expression at $r \approx d/2$, so

$$\operatorname{codim} I \sim \frac{3d^2}{4}.$$

Since the ambient dimension is $2d^2$, the incidence locus occupies codimension about 37.5% of the ambient dimension.

For the tested dimensions:

d	ambient dimension	codim I	percentage
2	8	3	37.5%
3	18	7	38.9%
4	32	12	37.5%
6	72	27	37.5%
10	200	75	37.5%

Corollary 3.3 (Rank-Protected Type IV Exclusion)**.**

If A has full column rank, then $\ker A = 0$ and $AB = 0$ implies $B = 0$. Therefore Type IV cannot occur on full-column-rank block strata. Dually, if B has full row rank, then $AB = 0$ implies $A = 0$.

Thus Type IV is structurally impossible on rank-protected projected block strata – namely when A has full column rank or B has full row rank – but constructible on rank-deficient sectorized matrix systems.

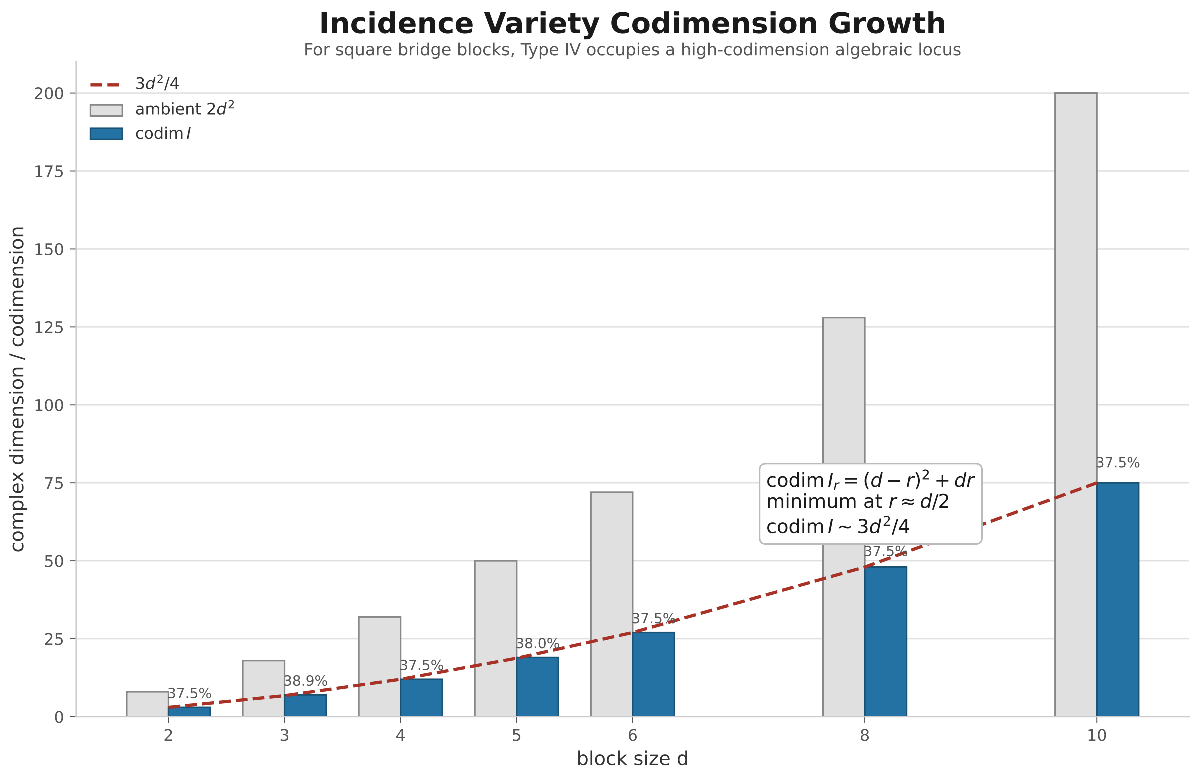

Figure 2. Incidence codimension growth. For square blocks, the dominant Type IV incidence locus has codimension asymptotic to $3d^2/4$, placing it in a high-codimension region rather than in the generic block stratum.

4 Generic Completion Roadmap

The codimension theorem turns Type IV from an unexplained obstruction into a geometric boundary. It is a proper algebraic degeneracy in the space of bridge products.

Remark 4.1 (Generic Completion Principle).

Outside a high-codimension incidence variety, rank-protected bridge products generically survive. Consequently, the observable pair (R_1, R_2) is expected to determine the first-depth invariant D.

This principle is the conceptual bridge from Paper V to Paper VII. Paper V shows why R_1 alone fails and identifies Type IV as the hard local exception. Paper VII identifies that exception as an incidence variety of quadratic codimension growth and formulates the generic completion statement away from it.

The resulting theorem layer separates into three steps:

$$\text{rank protection} \quad \Longrightarrow \quad \text{generic nonincidence} \quad \Longrightarrow \quad \text{completion away from incidence.}$$

The first two steps are algebraic consequences of the incidence calculation. The third step is the remaining completion conjecture.

Corollary 4.2 (Rank-Protected Product Survival)**.**

Let

$$A \in \mathbb{C}^{m\times n}, \qquad B \in \mathbb{C}^{n\times p}.$$

If A has full column rank and $B \neq 0$, then $AB \neq 0$. Dually, if B has full row rank and $A \neq 0$, then $AB \neq 0$.

Proof. *If A has full column rank, then $\ker A = 0$. Hence $AB = 0$ forces every column of B to lie in $\ker A$, so $B = 0$. The dual statement follows from applying the same argument to adjoints.*

Proposition 4.3 (Generic Nonincidence)**.**

Fix block dimensions (m, n, p). The Type IV bridge-failure locus

$$I = \{(A, B) : AB = 0,\ A \neq 0,\ B \neq 0\} \subset \mathbb{C}^{m\times n} \times \mathbb{C}^{n\times p}$$

is contained in a finite union of proper algebraic strata. On the stratum $\operatorname{rank} A = r$, its codimension is

$$\operatorname{codim} I_r = (m - r)(n - r) + pr.$$

In particular, for square block scale $m = n = p = d$, the dominant incidence codimension grows quadratically, asymptotic to $3d^2/4$.

Proof. *This is Theorem 3.1 applied over all admissible ranks. In the square case used for the asymptotic, the Type IV strata have $1 \le r \le d - 1$, excluding $r = 0$ and the rank-protected case $r = d$. In a general rectangular system, full row rank of A is not by itself rank-protecting when $m < n$; nonzero B may still land in $\ker A$. The rank-protected exclusions are exactly those in Corollary 3.3. The square-block asymptotic is Corollary 3.2.*

4.1 Conjecture 1 (Completion Away from Incidence)

Let $(V, \{Q_i\}, \{X_g\})$ be a sectorized observable framework satisfying the usual nondegeneracy and commutant-richness hypotheses needed to repair Type I and Type III obstructions. Assume further that every relevant bridge product avoids the Type IV incidence variety:

$$Q_i X_g Q_k X_h Q_j \neq 0$$

whenever the two factors are nonzero and the product is required as a bridge candidate.

Then the accessibility jet $\mathscr{J}_{\mathrm{acc}}$ determines the accessibility depth matrix D. Equivalently, under these richness and nondegeneracy hypotheses, on the generic non-incidence locus the discrete shadow of $\mathscr{J}_{\mathrm{acc}}$ is complete for first-depth accessibility.

Equivalently, under these hypotheses, the observable pair (R_1, R_2) together with the first-depth shadow of $\mathscr{J}_{\mathrm{acc}}$ determines D.

4.2 Interpretation

The conjecture separates the fate of exceptional mechanisms. Type III cancellation is a soft exceptional locus, repairable by higher-depth fields. Type IV incidence is a hard exceptional locus, but it has high codimension and measure zero in the ambient Lebesgue, equivalently Zariski-generic, matrix-pair space.

Thus generic completion does not say that exceptions do not exist. It says that the only hard local obstruction is algebraically nongeneric.

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5 Relation to Paper VI

Paper VI studies moving sectorized systems on normal spectral charts $\Sigma_{\mathrm{spec}} \subseteq \Sigma_{\mathrm{comm}}$. The accessibility jet varies as a smooth matrix-field object, while R_1, R_2, and D are discrete projections of it. The accessibility wall hierarchy is

$$\widehat{\Sigma}_{R_1} \subseteq \widehat{\Sigma}_{R_2} \subseteq \widehat{\Sigma}_D = \Sigma_{\mathrm{access}} \subseteq \Sigma_{\mathrm{spec}} \subseteq \Sigma_{\mathrm{comm}}.$$

Paper VII refines the residual R_2/D boundary by identifying the Type IV component:

$$\Sigma_{\mathrm{IV}} \subseteq \Sigma^{\circ}_{R_2} \cup \Sigma^{\circ}_{D}.$$

Type III and Type IV are not Paper VI wall categories. They are local mechanisms that can occur inside the static fibers over those walls. Paper VI answers where the observables jump in moduli space; Paper VII asks when the accessibility jet has enough information to complete the depth calculation in one fixed sectorized system.

This gives the closing line: Paper V says that exceptional mechanisms exist; Paper VI says that exceptional mechanisms move on the moduli space; Paper VII says that, generically, exceptional mechanisms either repair or disappear.

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6 Computational Proposition

The computational support for the completion theory is concentrated in three Paper VII support scripts, abbreviated below by role:

Label	Role
VII-A atlas	Type III/IV synthetic systems, represented-atlas limitation, exact $(R_1, R_2) \to D$ hash audit
VII-B codimension	incidence codimension table, random-pair check, perturbation instability
VII-C bridge audit	bridge-level rank-protection and nonincidence audit for Rubik, synthetic, and random systems

The corresponding repository scripts are `atlas_r2_boundary.py`, `incidence_variety_codim.py`, and `rank_protected_bridge_audit.py` in `experiments/paper7/`.

The stable support table is organized by claim-status gates:

Claim supported	Source	Observed result	Status
Rank-protected bridge survival	VII-C	Corollary 4.2 checked for constructed rank-protected blocks in dimensions $2, 3, 4$; no violation	theorem-support gate
Type IV incidence has high codimension	VII-B	square-block codimension is asymptotic to $3d^2/4$; tested values give about 37.5% of ambient dimension	theorem-support computation
Random matrix pairs avoid $AB = 0$	VII-B	0/400000 tested nonzero square matrix pairs satisfy $AB = 0$	empirical sanity check
Type IV incidence is perturbatively unstable	VII-B	perturbation at $\varepsilon = 10^{-6}$ breaks constructed $AB = 0$ in 100/100 trials for $d = 2, 3, 4$	empirical stability check
Tested random bridge products avoid incidence	VII-C	five fixed-seed random systems give 1080/1080 safe bridge products and zero incidence candidates	computational evidence
Type III/IV boundary is sharp at bridge level	VII-C	synthetic Type III has 4/4 rank-protected bridges; synthetic Type IV has 4/4 incidence candidates	constructed boundary check
Rubik is structured rather than random-generic	VII-C	Rubik has 2376 bridge products, 1848 generic nonincidence products, and 528 bridge-level incidence candidates concentrated in four sector triples	structured-laboratory evidence
Tested ablations keep missing incidence	VII-C	dimension 6–16, rank-deficient fraction 0–0.8, generator count 2–6: 75 random runs, 23400 audited bridge products, zero incidence candidates	exploratory only
Type III and Type IV separate the candidate class	VII-A	synthetic Type III lies in C; synthetic Type IV lies outside C	constructed examples
Exact (R_1, R_2) signatures showed no D disagreement	VII-A	80 systems produced 4 exact-hash classes with identical (R_1, R_2); all had identical D structure	computational evidence
Represented atlas is currently diagnostic	VII-A	regular-representation sectorizations produced nearly zero R_1 in most systems and no Type IV example	limitation, not negative theorem

6.1 Computational Proposition 4.3 (VII-A Completion Evidence)

The Paper VII support suite verifies the following finite statements.

(i) Type III versus Type IV. A synthetic Type III cancellation system lies inside the candidate completeness class C. A synthetic Type IV incidence system lies outside C.

(ii) Incidence instability. Starting from constructed Type IV systems, generic perturbation at $\varepsilon = 10^{-6}$ breaks $AB = 0$ in 100/100 trials for $d = 2, 3, 4$.

(iii) Random-pair absence. Across 400000 random square matrix pairs in the tested dimensions, no nonzero pair satisfied $AB = 0$.

(iv) Exact signature audit. In 80 diverse random sectorized systems, there were 4 exact-hash equivalence classes with identical R_1 and R_2 arrays. All 4 classes had identical $D_{\max}$ and identical per-depth structure. No D disagreement was found.

(v) Claim-status-gated bridge audit. The rank-protected bridge audit is split into five gates. Gate A is theorem-support: Corollary 4.2 was checked on constructed rank-protected blocks in dimensions $2, 3, 4$, with no violation. Gate B checks the Type III/IV boundary: the synthetic Type III model has 4/4 rank-protected bridge products and zero incidence candidates, while the synthetic Type IV model has 4/4 incidence candidates. Gate C is computational evidence for generic nonincidence: five fixed-seed random systems $(42, \ldots, 46)$ give 1080/1080 safe bridge products and zero incidence candidates.

Gate D is diagnostic. The Rubik system behaves differently: among 2376 bridge products, 1848 are generic nonincidence products and 528 are bridge-level incidence candidates. These are candidate products in the audit, not certified Type IV accessibility obstructions. They are concentrated in four sector triples, in zero-based sector indices $(2, 6, 8)$, $(5, 6, 8)$, $(8, 6, 2)$, and $(8, 6, 5)$.

Equivalently, in the RIME sector labels, they involve the triples $(S3, S7, S9)$ and $(S6, S7, S9)$ together with their reversed orientations.

Gate E is exploratory. It varies total dimension 6–16, rank-deficient fraction 0–0.8, and generator count 2–6. Across 75 random runs and 23400 audited bridge products, no incidence candidate appears. This ablation is evidence for the genericity picture, not a theorem.

These computations do not prove Conjecture 1. They support the claim that Type IV is the hard boundary and that, away from this boundary and under the tested nondegeneracy/richness conditions, the accessibility jet behaves as a complete object in the tested families.

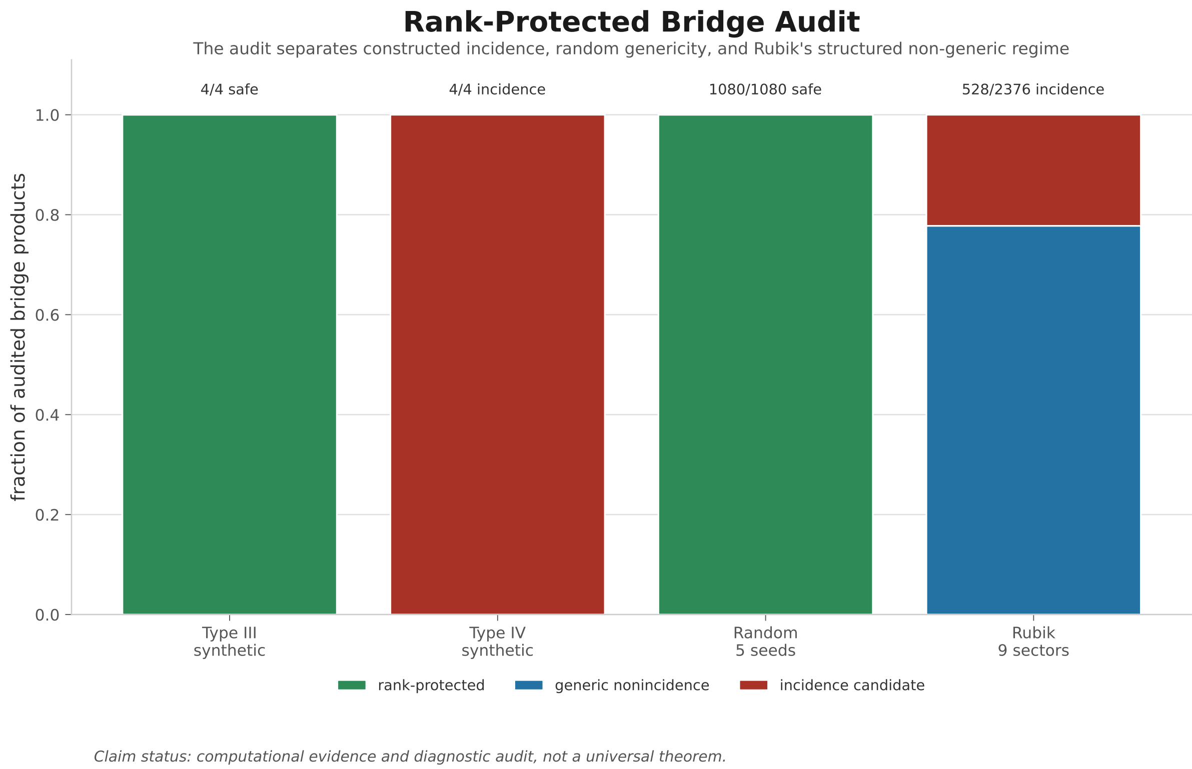

Figure 3. Rank-protected bridge audit. The tested random families show no audited incidence candidates; the synthetic Type IV boundary is sharp; Rubik concentrates bridge-level incidence candidates, not certified Type IV accessibility obstructions, in a small structured subset of sector triples.

The Rubik bridge audit should not be read as a contradiction to generic nonincidence. It shows that Rubik lies in a structured, rank-deficient region of the sectorized-system space. This is precisely why the Rubik laboratory is useful: it is rich enough to expose nongeneric incidence geometry that random systems did not hit in the tested audits.

Remark 6.1 (Transport-to-Incidence Hub Recurrence).

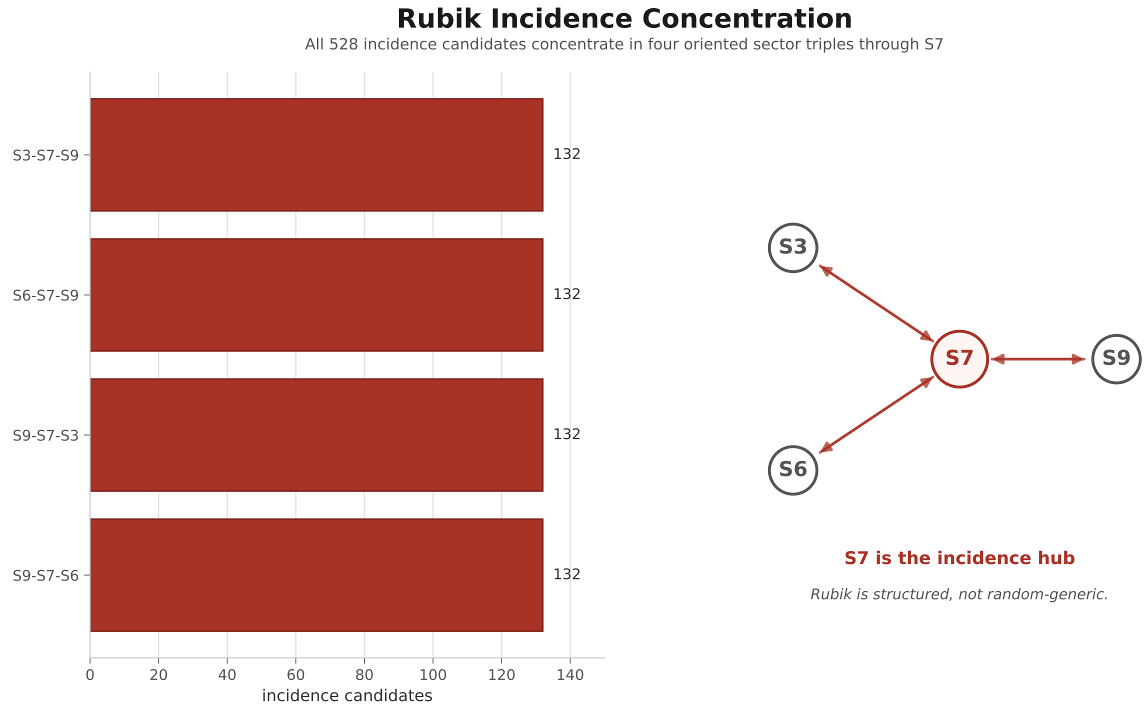

Figure 4. Rubik incidence concentration. In the bridge audit, all observed bridge-level incidence candidates cluster in four oriented sector triples through the $S7$ hub. This locates Rubik in a structured exceptional region rather than in the tested random-generic regime.

The four Rubik incidence candidate triples are $(S3, S7, S9)$ and $(S6, S7, S9)$, together with their reversed orientations. Thus $S7$ is the bridge-level incidence hub in the Paper VII audit. This echoes the Paper II transport geometry, where $S7$ lies at the end of the $S5$–$S6$–$S7$ transport chain and next to the unique Type II edge $S8$–$S9$ [4]. No theorem-level relation between the transport hub and the incidence hub is claimed here. Hub recurrence may indicate a later notion of sector centrality linking transport, incidence, and accessibility.

6.2 Represented Atlas Limitation

The represented-atlas component is currently diagnostic rather than decisive. Regular-representation sectorizations produced nearly zero R_1 in most tested systems, meaning they do not yet probe rich cross-sector Type IV behavior. This is a known limitation of the current atlas.

The lesson is useful: Type IV requires mixed sectorizations where cross-sector blocks exist and products nevertheless vanish by incidence. Regular representation sectorizations tend to suppress the first condition.

The rank-protected bridge audit adds the opposite lesson. Tested random sectorized systems miss the incidence variety in the audited bridge products, whereas Rubik hits it in a structured and repeatable way. Thus Rubik is not a generic point of the sectorized-system space. It is a stable carrier of high-codimension algebraic structure, which is exactly why it functions as a useful laboratory for the RIME program.

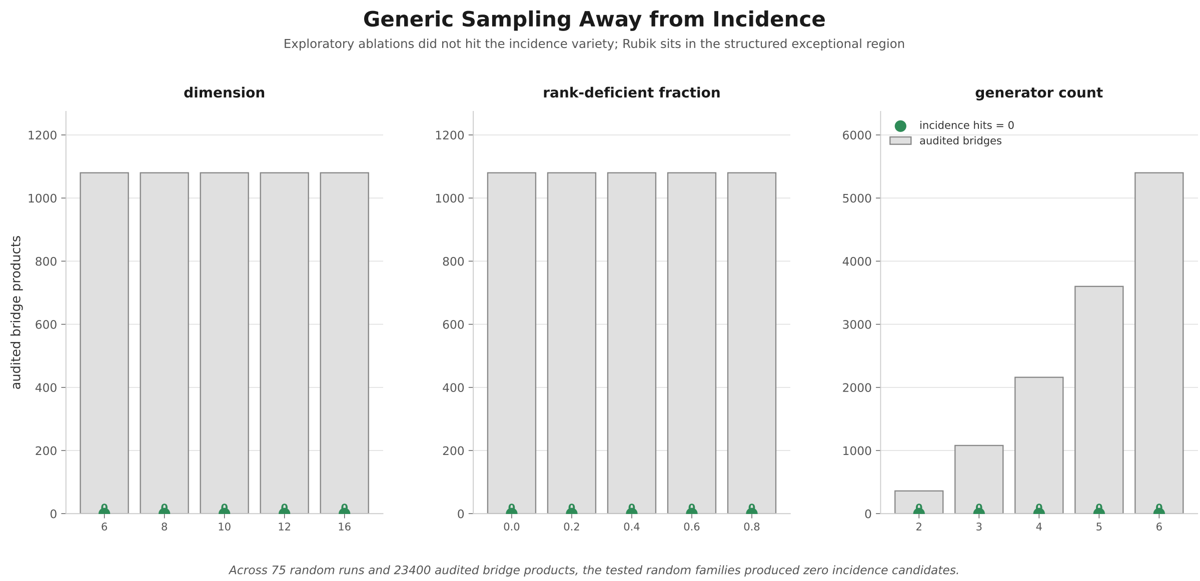

Figure 5. Generic sampling landscape. Across the audited products in the tested random families, incidence candidates remain absent. The final panel points back to Paper VI: structured exceptional carriers require a moduli-space and wall-crossing description.

7 Scope and Open Problems

7.1 What This Paper Establishes

- Type IV incidence is the algebraic variety $I = \{(A, B) : AB = 0,\ A, B \neq 0\}$.
- On fixed-rank strata, $\operatorname{codim} I_r = (m - r)(n - r) + pr$.
- For square blocks, the dominant incidence codimension is asymptotic to $3d^2/4$.
- Rank-protected projected block strata exclude Type IV.
- Random bridge products in the tested families are fully safe from incidence, while Rubik exhibits a structured incidence sublocus concentrated in a small set of sector triples.
- Computational evidence separates Type III from Type IV and supports generic completion away from incidence.

7.2 What This Paper Does Not Establish

- It does not prove a universal theorem that (R_1, R_2) determines D.
- It does not prove that all Type IV incidence conditions remain stable under arbitrary Lie bracketing at all depths.
- It does not classify higher-depth walls $R_3, R_4, \ldots$.
- It does not yet construct a rich representation-derived Type IV accessibility obstruction; Rubik currently supplies bridge-level incidence candidates.

7.3 Open Problems

1. **Generic completion theorem.** Prove Conjecture 1 or find the minimal non-incidence counterexample.

2. **Algebraic richness.** Express the commutant-richness condition needed to repair Type III cancellations in terms of the representation and the accessibility jet.

3. **Incidence in represented systems.** Determine the intersection of Σ_{IV} with representation-derived sectorization loci.

4. **Higher-depth completion.** Extend the incidence/cancellation analysis from length-two witnesses to higher Hall layers.

Appendix A Perturbation Check

The Type IV boundary is perturbatively unstable in the tested models.

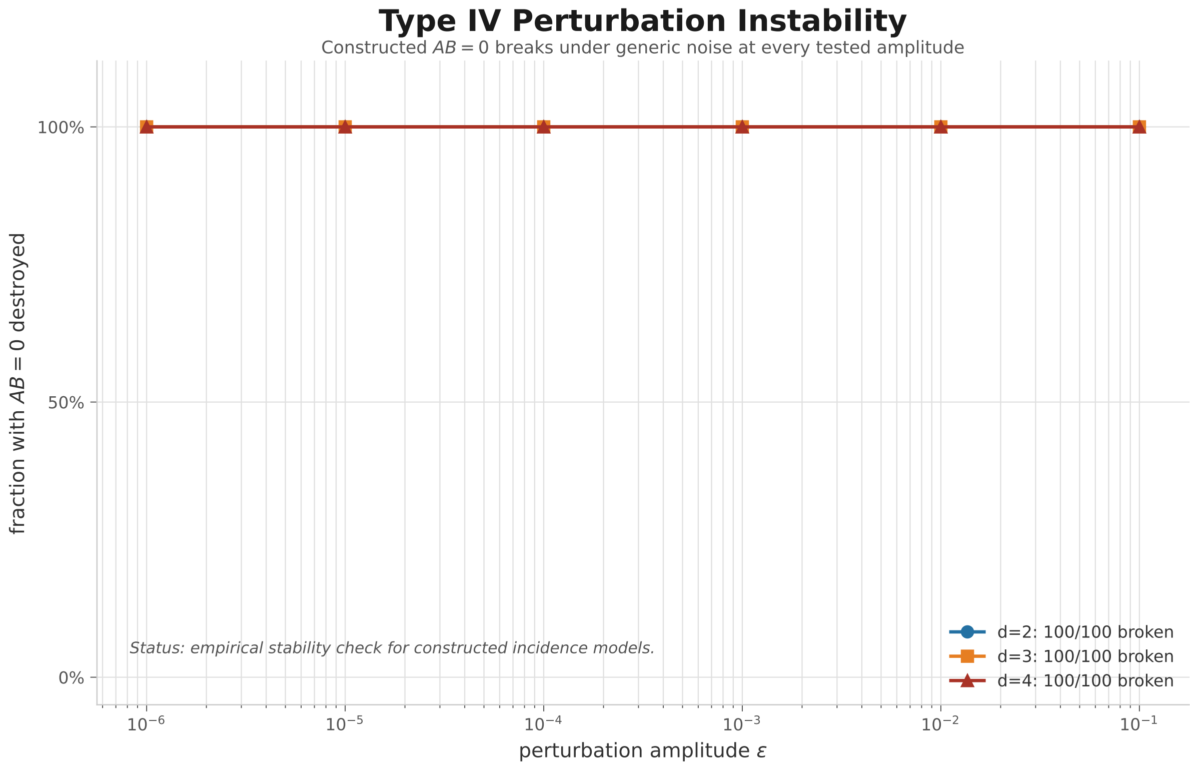

Figure 6. Type IV perturbation instability. In the synthetic incidence families, every tested perturbation amplitude breaks the constructed $AB = 0$ relation.

Appendix B Claim-Status Gates

The support suite is intentionally tiered: theorem support, computational evidence, diagnostics, and exploratory scans are kept distinct.

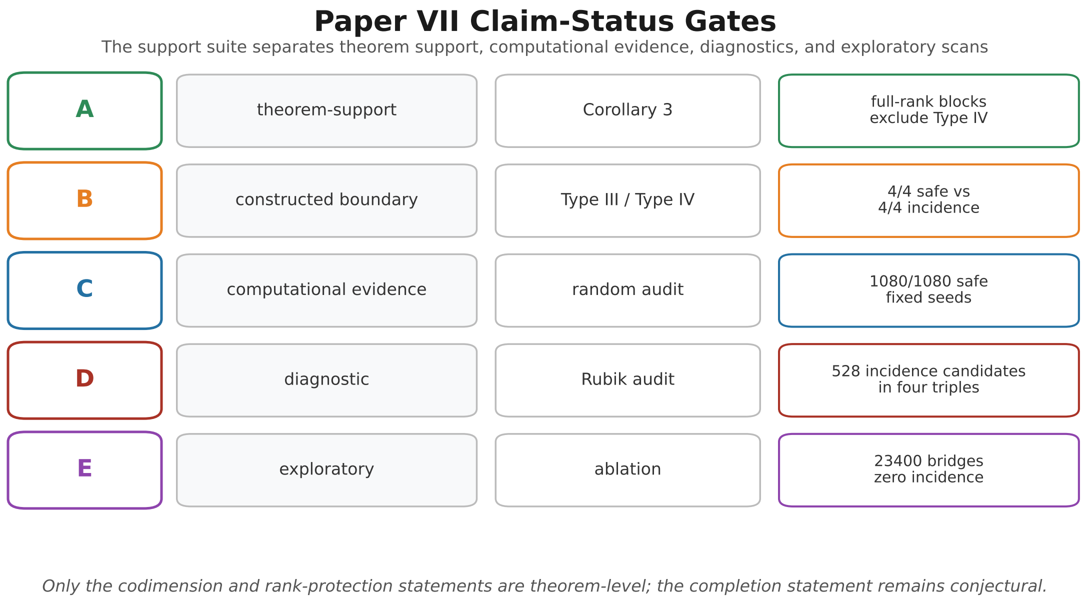

Figure 7. Paper VII claim-status gates. The codimension and rank-protection statements are theorem-level; random-family audits are computational evidence; the completion principle remains conjectural.

Appendix C Cross-Species SOF Diagnostics

The main body of this paper is about generic completion and the Type IV incidence boundary. The following diagnostics are not part of the proof of Conjecture 1. They record a separate sanity check: the observables R_1, R_2, and D are not specific to the Rubik representation.

C.1 Note on the Sectorized Observable Framework (SOF)

The objects appearing throughout Papers V–VII naturally admit a common description, which we refer to as a Sectorized Observable Framework (SOF). We describe an SOF by

$$\mathcal{F} = (V, \{Q_i\}, \mathcal{X}),$$

where V is a finite-dimensional representation space, $\{Q_i\}$ is a distinguished family of sector projectors, and $\mathcal{X}$ is a chosen observable family, typically generators, transfer operators, or related observables.

Throughout this appendix we use the term "Sectorized Observable Framework (SOF)" as a neutral organizational language for the sectorized systems appearing in the RIME program. It is intended as terminology only; no additional mathematical structure beyond $(V, \{Q_i\}, \mathcal{X})$ is assumed here.

The quantities R_1, R_2, and D are defined relative to this triple rather than to any particular Rubik representation. The Rubik cube, quantum gate systems, Markov systems, and graph systems considered below are therefore interpreted as different realizations of the same framework.

The unifying object is not a particular wall theory, but the sectorization. Different systems may share the same sectorized observable language while exhibiting different deformation geometries.

A systematic development of the SOF language lies beyond the scope of the present paper and is left for future work.

C.2 Quantum Gate Systems

For the quantum diagnostic, $V = (\mathbb{C}^2)^{\otimes q}$, the sectors are computational-basis projectors $|b\rangle\langle b|$, and the operator family is obtained from skew-Hermitian logarithmic gate generators. The audit uses the same convention as the main paper: R_1 records projected generator support, R_2 records projected commutator survival, and D records first Lie-depth accessibility up to the tested depth.

The diagnostic table is:

Gate set	q	R_1 offdiag	R_2 offdiag	frozen R_1	frozen D	D-repaired	$D_{\max}$
Pauli $\{X, Z\}$	2	16.7%	33.3%	8	8	0	∞
Pauli $\{X, Y, Z\}$	2	22.2%	22.2%	8	8	0	∞
Clifford $\{H, S, \mathrm{CNOT}\}$	2	16.7%	33.3%	6	0	6	2
Clifford $\{H, S, \mathrm{CZ}\}$	2	11.1%	16.7%	8	8	0	∞
Universal $\{H, T, \mathrm{CNOT}\}$	2	16.7%	33.3%	6	0	6	2
Pauli $\{X, Z\}$	3	7.1%	14.3%	48	48	0	∞
Clifford $\{H, S, \mathrm{CNOT}\}$	3	7.1%	14.3%	44	32	12	∞
Universal $\{H, T, \mathrm{CNOT}\}$	3	7.1%	14.3%	44	32	12	∞

Here ∞ means unreached within the tested finite Lie-depth cutoff.

The observations are:

Observation C.1 (Entangling generators create additional accessibility channels). In the tested computational-basis sectorization, CNOT opens channels that are absent in product-only Pauli systems and in the tested CZ variant. The ordering observed in the diagnostic is

$$\mathrm{CNOT} > \mathrm{CZ} > \text{product-only}$$

with respect to D-repair in the tested gate families.

Observation C.2 (Lie-depth repairs accessibility beyond first-order transport). R_1-frozen does not imply D-frozen. In the two-qubit Clifford+CNOT and Universal+CNOT systems, 6 sector pairs frozen at the first support layer become accessible at higher Lie depth. In the three-qubit versions, 12 such repairs are observed within the tested depth range.

Observation C.3 (Universality alone does not enrich low-order accessibility). Adding the T gate does not change the audited $R_1/R_2/D$ summary relative to Clifford+CNOT in the tested 2- and 3-qubit systems. Thus computational universality of the gate set and low-order sector accessibility are different notions.

C.3 Markov and Graph Systems

Markov and graph systems provide a second diagnostic layer. For Markov systems, V is a finite state space, sectors are state projectors, and the operator is a rate or logarithmic transition operator. For graph systems, V is a vertex space, sectors are vertex or spectral sectors, and operators may include adjacency, Laplacian, walk, or directed-edge data.

The current diagnostic results are:

System	Species	Sectors	Generators	R_1 offdiag	frozen R_1	frozen D	D-repaired	$D_{\max}$
Markov chain	Markov	3	1	100.0%	0	0	0	0
Absorbing Markov	Markov	3	1	66.7%	2	2	0	∞
Graph K_3	Graph	3	2	50.0%	0	0	0	0
Graph P_3	Graph	3	2	33.3%	2	2	0	∞
Graph C_4	Graph	4	2	33.3%	4	4	0	∞

These examples show that the audit interface is portable, but they do not establish a generic completion theorem. Complete or strongly connected examples may have $D = 0$ without any repair phenomenon, while sparse examples may remain frozen for the chosen operator family.

Unlike Rubik or CNOT-generated systems, these examples employ only a single effective transport generator or an already-complete connectivity pattern, leaving no opportunity for higher-order Lie repair.

C.4 Cross-Species Observations

Across the quantum, Markov, and graph diagnostics, the same audit interface applies once the sectorized data $(V, \{Q_i\}, \mathcal{X})$ are fixed. The results support the SOF hypothesis that R_1, R_2, and D are sectorized observables rather than Rubik-specific quantities.

The diagnostics also separate three regimes:

1. product-only or single-effective-generator systems, where higher-order Lie repair is absent in the tested range;
2. already-complete systems, where $D = 0$ leaves no repair problem;
3. sufficiently rich noncommuting systems, such as CNOT-generated examples, where R_1-frozen pairs can be repaired at higher Lie depth.

C.5 Empirical SOF Principle

Across the tested SOF species, nontrivial D-repair has appeared only in systems equipped with sufficiently rich noncommuting transport generators, visible to the chosen sectorization. Tested single-generator or effectively degenerate systems either are already connected at first depth, or remain frozen in the tested Lie filtration.

This principle is empirical. It should be read as a routing rule for future SOF examples, not as a theorem. To promote it, one would need a precise definition of sufficient noncommutative transport richness and a proof that it is necessary or generic for D-repair.

The scripts supporting this appendix are the quantum SOF audit and the Markov/graph SOF audit in `experiments/quantum/`.

Bibliographic Notes

Program lineage. Paper VII depends directly on Papers IV–VI. Paper IV supplies the fixed-arrangement collision quotient [3]; Paper V supplies the local length-2 repair calculus and the Type III/IV mechanism taxonomy [2]; Paper VI supplies the deformation geometry of the commutativity locus, accessibility jets, and accessibility walls [5]. Thus Paper VII is the static completion layer after collision geometry, repair calculus, and wall geometry have been separated.

External background. The matrix and rank arguments use standard finite-dimensional linear algebra [9]. The representation-theoretic background is standard finite-group representation theory [12], while the generic-accessibility vocabulary is anchored in geometric control theory [10, 1]. The Type IV boundary is an incidence/Schubert condition, so the algebraic-geometry background is the usual one for determinantal and incidence loci [6, 8]. Hall words and weighted Lie-depth language continue the free-Lie-algebra framework already used in Paper V [7, 11]. Paper VII does not use the association-scheme quotient story as its main background; that belongs with the fixed collision geometry of Paper IV.

Computational provenance. The support suite is organized as the VII-A atlas experiment, the VII-B incidence codimension experiment, and the VII-C rank-protected bridge audit. The manuscript-level reproducibility notes remain attached to the support scripts in `experiments/paper7/`, while the repository stores the full logs and tables used to check the claim-status gates.

Program status. Paper V identifies local exceptional mechanisms. Paper VI places those mechanisms on the generator-set moduli space. Paper VII identifies the hard mechanism, Type IV incidence, as a high-codimension algebraic degeneration. Within the SOF language $\mathcal{S} = (V, \{Q_i\}, \mathcal{X})$, Type IV is an incidence stratum in the block geometry of $\mathcal{S}$, not a Rubik anomaly. The resulting completion theory is the current static closure framework: under the stated richness and nondegeneracy hypotheses, and outside the relevant Type IV incidence variety, the Generic Completion Principle predicts that (R_1, R_2) determines the first-depth invariant D. Deformation of the exceptional incidence locus belongs to the moduli-space line opened by Paper VI.

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